# String

String – String represents group of characters. Strings are enclosed in double quotes or single quotes. The *str* data type represents a String.

Ex:- “Hello”, “GeekyShows”, ‘Rahul’

str1 = “GeekyShows”

# Creating String

Single Quotes

str1 = ‘GeekyShows’

Double quotes

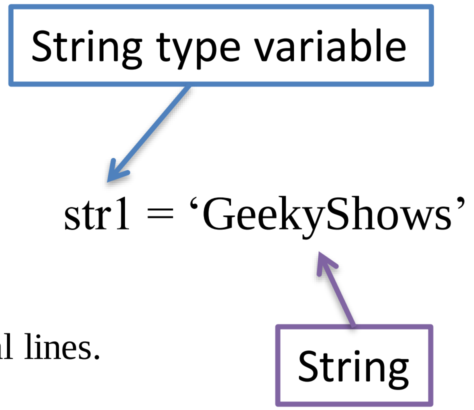
str2 = “GeekyShows”

Triple Single Quotes – This is useful to create strings which span into several lines.

str3 = ‘‘‘ Hello Guys
Please Subscribe
Geekyshows’’’

Triple Double Quotes – This is useful to create strings which span into several lines.

str4 = “““Hello Guys
Please Subscribe
Geekyshows ”””



# Creating String

Double Quote inside Single Quotes

str5 = ‘Hello “Geeky Shows” How are you’

Single Quote inside Double quotes

str6 = “Hello ‘Geeky Shows’ How are you”

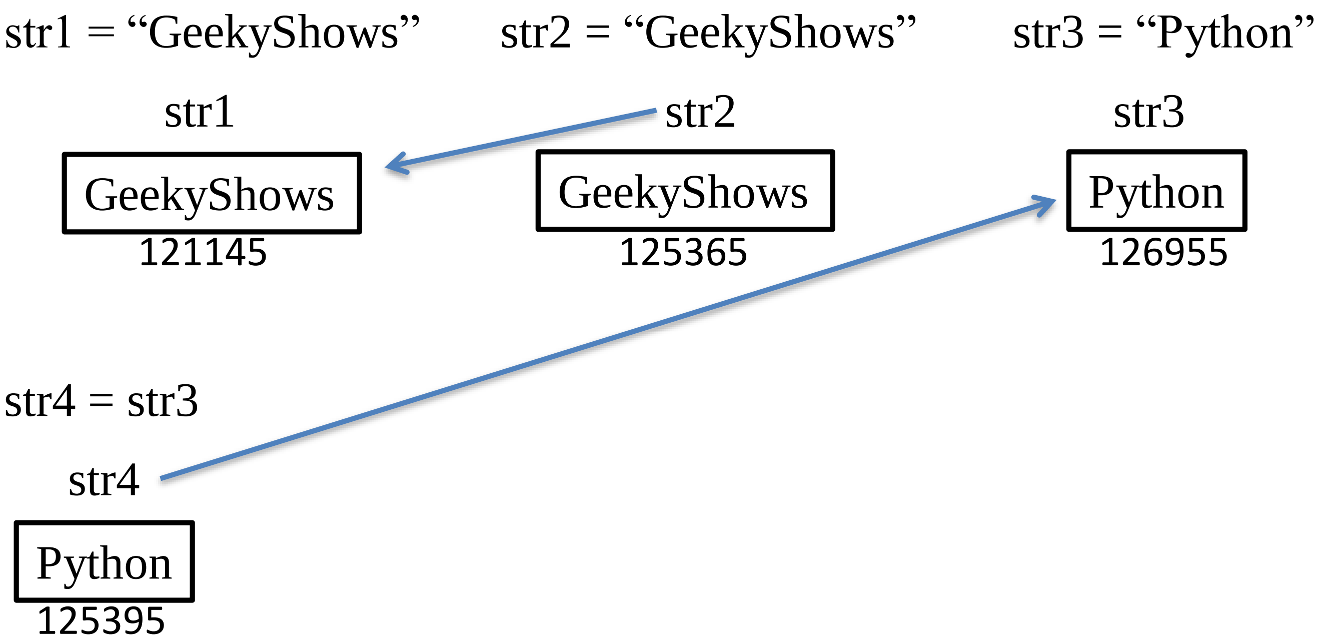
Using Escape Characters

str7 = “Hello \nHow are You ?”

Raw String – Row string is used to nullify the effect of escape characters.

str8 = r“Hello \nHow are You ?”

# Memory Allocation



# Memory Allocation

str1 = “GeekyShows”

str1

GeekyShows

121145

str1 = “Python”

str1

Python

125365

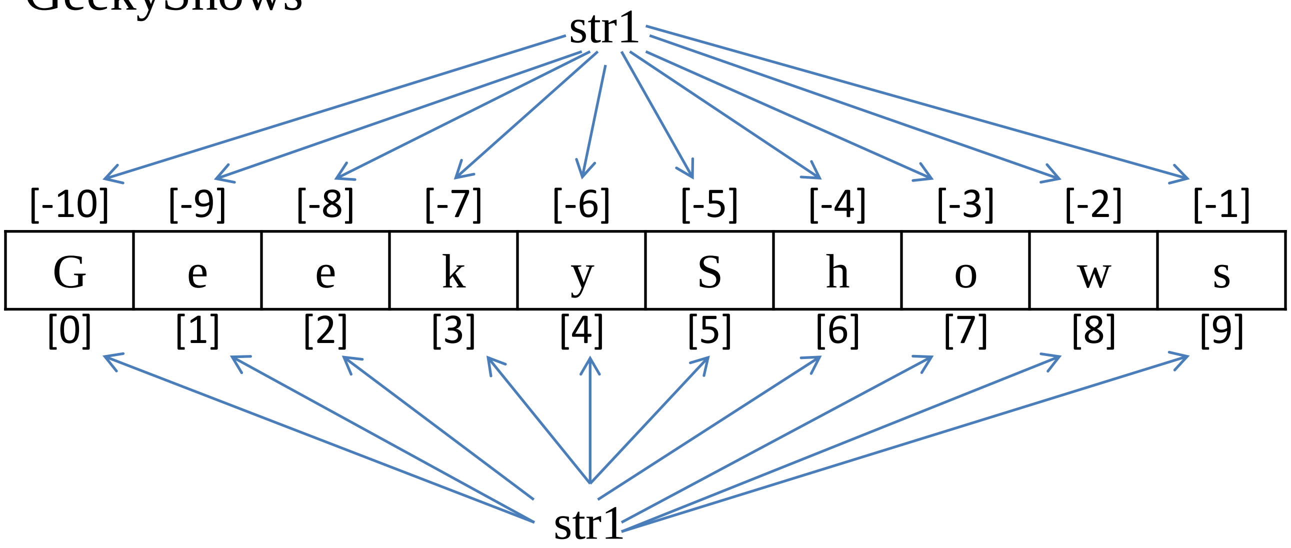
Since value “GeekyShows” becomes unreferenced object, it is removed by garbage collector.

# Index

Index represents the position number of characters in a string.

Ex:-

str1 = “GeekyShows”



# Accessing Character and String

str1 = “GeekyShows”

| [-10] | [-9] | [-8] | [-7] | [-6] | [-5] | [-4] | [-3] | [-2] | [-1] |
|---|---|---|---|---|---|---|---|---|---|
| G | e | e | k | y | S | h | o | w | s |
| [0] | [1] | [2] | [3] | [4] | [5] | [6] | [7] | [8] | [9] |

print(str1[0])

print(str1[1])

print(str1[2])

print(str1[3])

print(str1[-1])

print(str1)

# String Length

Length of string represents the number of characters in a string.

len( ) Function is used to get length of string.

Ex:-

str1 = “GeekyShows”

n = len(str1)

# Access using Loop

```
str1 = “GeekyShows”
for c in str1:
	print(c)

for i range(len(str1)):
	print(str1[i])

i = 0
while i < len(str1) :
	print(str1[i])
	i+=1
```

# Mutable and Immutable Object

- Mutable Object – Mutable objects are those object whose value or content can be changed as and when required.

  Ex:- List, Set, Dictionaries

- Immutable Object – Immutable objects are those object whose value or content can not be changed.

  Ex:- Numbers, String, Tuple

# Immutable String

In Python, Strings are immutable object which means it’s value or content can not be changed.

str1 = “GeekyShows”

| [-10] | [-9] | [-8] | [-7] | [-6] | [-5] | [-4] | [-3] | [-2] | [-1] |
|---|---|---|---|---|---|---|---|---|---|
| G | e | e | k | y | S | h | o | w | s |
| [0] | [1] | [2] | [3] | [4] | [5] | [6] | [7] | [8] | [9] |

str1[4] =“i”

TypeError

# Slicing String

Slicing on String can be used to retrieve a piece of the string. Slicing is useful to retrieve a range of elements.

Syntax:-

new_string_name = string_name[start:stop:stepsize]

- start – It represents starting point. By default its 0
- stop – It represents ending point.
- stepsize – It represents step interval. By default It is 1
- If start and stop are not specified then slicing is done from $0^{th}$ to $n\text{-}1^{th}$ elements.
- Start and Stop can be negative number.

# Repetition Operator

Repetition operator is used to repeat the string for several times. It is denoted by *

Ex:-

“$” * 7

str1 = “Geeky Shows ”

str1 * 5

# Concatenation Operator

Concatenation operator is used to join two string. It is denoted by +

Ex:-

“Geeky” + “Shows”

str1 = “Geeky ”

str2 = “Shows”

str1 + str2

# Comparing String

str1 = “GeekyShows”
str2 = “GeekyShows”
result = str1 == str2

str1 = “GeekyShows”
str2 = “Python”
result = str1 == str2

str1 = “A”
str2 = “B”
result = str1 < str2

| Operators | Meaning |
|---|---|
| < | Less than |
| > | Greater than |
| <= | Less than or equal to |
| >= | Greater than or equal to |
| == | Equal to |
| != | Not equal to |